Probability Theory for Quants

Mathematical Framework for Risk, Uncertainty, and Market Dynamics
Measures, Martingales, and Market Regimes Made Visual

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January 28, 2026

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Common employers / interview contexts for this material

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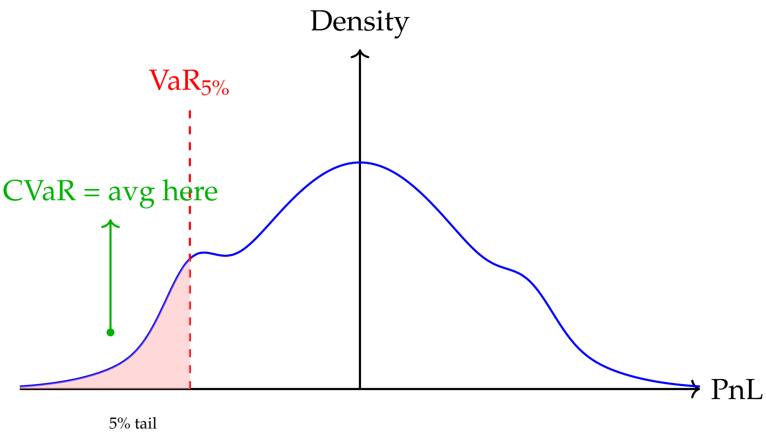

Figure 3: VaR is the quantile; CVaR is the conditional expectation in the tail.

2.6 Spectral Risk Measures = Weighted Averages

CVaR weights all tail observations equally. Spectral risk measures generalize this:

Core Formula

$$\rho(X) = -\int_0^1 \phi(p) F_X^{-1}(p) dp$$

Where ϕ is a weighting function (risk spectrum).

CVaR: $\phi(p) = \frac{1}{\alpha}\mathbf{1}_{p<\alpha}$ (uniform on tail, zero elsewhere)
Exponential spectral: $\phi(p) = \frac{e^{-p/\gamma}}{\gamma(1-e^{-1/\gamma})}$ (more weight on extreme tail)

Practitioner Insight

Risk Aversion as Spectral Weight
Different spectral weights reflect different risk preferences: - Risk-neutral: $\phi = 1$ (uniform, just expectation) - VaR: $\phi = \delta_\alpha$ (Dirac at quantile) - CVaR: Uniform on $[0, \alpha]$ - Wang transform: $\phi(p) = \Phi(\Phi^{-1}(p) + \lambda)$ (distortion)
Interview: "I use spectral risk measures to embed risk aversion directly into the pricing kernel. The weight function ϕ is calibrated to the firm's utility function or regulatory requirements. This unifies pricing and risk management."

"Risk Aversion as Tilt" (Visualization)

Think of $\phi(p)$ as a seesaw: CVaR puts equal weight on the worst α%; exponential spectral puts more weight on the extreme left (deep tail). **Quick reconstruction:** If you forget the formula, remember: $\rho = -\int_0^1 \phi(p) F^{-1}(p) dp$ is just a weighted average of quantiles, where ϕ is your "fear function."

2.7 Convex Risk Measures and Penalty Functions

Beyond coherence (positive homogeneity), convex risk measures allow for liquidity effects:

Core Formula

$$\rho(X) = \sup_{\mathbb{Q} \in \mathcal{M}} \{\mathbb{E}^{\mathbb{Q}}[-X] - \alpha(\mathbb{Q})\}$$

Where $\alpha(\mathbb{Q})$ is a penalty for using alternative measures.

5.1 Martingale Definition = Fair Game

A process M_t is a martingale if:

Core Formula

$$\mathbb{E}[M_t|\mathcal{F}_s] = M_s \quad \text{for } s \leq t$$

Conditions (typical): M_t is adapted and integrable (e.g., $\mathbb{E}[|M_t|] < \infty$ for each t).

"Fair Game = Frozen Expectation" (Mnemonic)

Martingale: Future expectation is *frozen* at today's value. **Submartingale:** "Sub" as in "submarine going up" (drift up, favorable). **Supermartingale:** "Super" as in "superman flying away" (drift down, unfavorable). **Check:** Under $\mathbb{Q}$, discounted stock is martingale; under $\mathbb{P}$, it's usually submartingale (equity risk premium).

Properties: - Constant expectation: $\mathbb{E}[M_t] = \mathbb{E}[M_0]$ - Driftless: No predictable trend - Represents "efficient market" in weak form

"Martingale = No Free Lunch"

If you knew $\mathbb{E}[S_{t+1}|\mathcal{F}_t] > S_t$, you would buy. Competition eliminates this. In efficient markets, discounted prices are martingales (under $\mathbb{Q}$).

5.2 Doob-Meyer Decomposition = Signal vs Noise

Any process X_t can be decomposed:

Core Formula

$$X_t = M_t + A_t$$

where: - M_t = Martingale (noise, unpredictable) - A_t = Predictable, finite variation (trend/signal)

Trading application: - A_t = Alpha (predictable drift) - M_t = Efficient market noise (untradeable pure risk)

Practitioner Insight

Alpha Extraction as Doob Decomposition
Your trading strategy attempts to separate A_t from M_t:

$$\text{Return}_t = \underbrace{\text{Alpha}}_{A_t} + \underbrace{\text{Beta/Market}}_{M_t}$$

Sharpe ratio depends on $A_t/\sigma(M_t)$. High Sharpe means large predictable component relative to noise.
Interview: "I decompose returns into martingale (unhedgeable) and drift (alpha). My models focus on predicting the drift component while managing exposure to the martingale risk. This is the Doob-Meyer decomposition applied to trading."

5.25 The Martingale Transform (Discrete Stochastic Integral)

If M_n is a martingale and H_n is **predictable** (known at $n-1$), then:

"Memoryless = State Sufficiency" (Problem Solving)

Test for Markov: If knowing the entire history $\{S_u\}_{u<t}$ gives no better prediction than knowing just S_t, it's Markov. **Asian option hack:** If asked to price Asian option (non-Markov) via PDE, immediately say: "I augment the state with $A_t = \int_0^t S_u du$. Now (S_t, A_t) is Markov."

6.2 Transition Kernels = Evolution Operators

Discrete state: Transition matrix P where $P_{ij} = \mathbb{P}(X_{n+1} = j | X_n = i)$.

Continuous state: Transition density $p(t, x; T, y)dy = \mathbb{P}(X_T \in dy | X_t = x)$.

Core Formula

$$\text{Chapman-Kolmogorov: } p(s, x; u, z) = \int p(s, x; t, y)p(t, y; u, z)dy$$

Interpretation: Multi-period transitions are compositions of single-period transitions.

Practitioner Insight

Credit Rating Transitions
Moody's transition matrix P (8x8 for ratings AAA to D):

$$P = \begin{bmatrix} 0.90 & 0.08 & 0.01 & \dots \\ 0.05 & 0.85 & 0.08 & \dots \\ \vdots & & & \ddots \end{bmatrix}$$

n-step ahead: Compute P^n. Probability of default from BB in 2 years = $(P^2)_{\text{BB,D}}$.
Interview: "I use the Chapman-Kolmogorov equation to compute multi-period default probabilities. The matrix P^n gives the exact distribution after n periods. For continuous time, I solve the Kolmogorov backward equation."

6.3 Generators $\mathcal{L}$ = Infinitesimal Description

For a diffusion $dX_t = \mu(X_t)dt + \sigma(X_t)dW_t$:

Core Formula

$$\mathcal{L}f(x) = \mu(x)f'(x) + \frac{1}{2}\sigma^2(x)f''(x)$$

"Generator = Infinitesimal P&L" (Interpretation)

When applying Itô's lemma, $df = \mathcal{L}f dt + \sigma f' dW$. **Mnemonic:** $\mathcal{L}f$ is your *expected* P&L per unit time (theta), $\sigma f'$ is your instantaneous risk (delta). **Quick calc:** For $f(S) = S^2$, $\mathcal{L}f = 2\mu S^2 + \sigma^2 S^2$ (drift + convexity).

Interpretation: $\mathcal{L}$ is the derivative of the semigroup: $\frac{d}{dt}\mathbb{E}[f(X_t)] = \mathcal{L}f$.

Connection to PDEs: Feynman-Kac formula links expectations to PDE solutions.